Basics of FIM: Statistical Learning Sample Questions

October 2026

Block 2: Statistical Learning

Task 1: Multiple Choice

Cross the correct answer. In case that you want to change your answer, color the box of the previous answer completely and cross the new correct answer.

- ■ Answer a).
- □ Answer b).
- ☒ Answer c).
- □ Answer d).

There are no penalty points.
Exactly one answer is correct.
For every correct answer you receive **1 point**.

1. In a statistical learning context, let $\hat{f} : X \to Y$ denote the estimated mapping from predictors to response. Which of the following statements describes the typical behavior as the flexibility of $\hat{f}$ increases?

   ☐ Its variance increases and its bias decreases.

   ☐ Its variance decreases and its bias increases.

   ☐ Its total error increases.

   ☐ Its total error decreases.

2. In a simple linear regression, given an explained sum of squares ESS of 8 and a residual sum of squares RSS of 12, what is the coefficient of determination R^2?

   ☐ 0.75.

   ☐ 0.60.

   ☐ 0.67.

   ☐ 0.40.

3. Let two features x^1 and x^2 be **dependent**, with realizations x^1_m ($m = 1, \ldots, k$) and x^2_s ($s = 1, \ldots, r$). Which of the following expresses their joint relative distribution correctly?

   ☐ $f(x^1_m, x^2_s) = f(x^1_m) + f(x^2_s | x^1 = x^1_m)$.

   ☐ $f(x^1_m, x^2_s) = f(x^1_m) \times f(x^2_s | x^1 = x^1_m)$.

   ☐ $f(x^1_m, x^2_s) = f(x^1_m) \times f(x^2_s)$.

   ☐ $f(x^1_m, x^2_s) = f(x^2_s | x^1 = x^1_m) / f(x^1_m)$.

Task 2: Python Programming

You fit a simple linear regression $y = \beta_0 + \beta_1 x$ by least squares to three observations. Below is a code snippet with missing parts.

```
1  xd = [1, 2, 3]
2  yd = [2, 4, 5]
3
4  xbar = sum(xd) / len(xd)
5  ybar = sum(yd) / len(yd)
6  sxy = sum((xi - xbar) * (yi - ybar) for xi, yi in zip(xd, yd)
       )
7  sxx = sum((xi - xbar) ** 2 for xi in xd)
8
9  b1_h = _______________ / _______________
10
11 b0_h = _______________ - _______________ * _______________
```

(a) Fill in the missing code in lines 9 and 11 such that `b1_h` and `b0_h` contain the least-squares estimates $\hat{\beta}_1$ and $\hat{\beta}_0$.

(b) State the two values that the `print` statement outputs.

```
print(b1_h, b0_h)
```

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Task 3: Python Programming

The lists `yt` and `yh` hold the observed responses and the fitted values of a linear regression model. The code below computes three sums of squares but uses uninformative variable names.

```
yt = [1.0, 3.0, 2.0, 4.0]
yh = [1.3, 2.1, 2.9, 3.7]

ybar = sum(yt) / len(yt)

A = sum((yi - ybar) ** 2 for yi in yt)
B = sum((yi - yhi) ** 2 for yi, yhi in zip(yt, yh))
C = sum((yhi - ybar) ** 2 for yhi in yh)

print(f'A = {A:.2f}, B = {B:.2f}, C = {C:.2f}, B + C = {B + C
    :.2f}')
```

(a) Assign each of `A`, `B` and `C` to one of the quantities TSS, RSS and ESS.

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(b) Spell out the abbreviation ESS and describe in one sentence what it quantifies about a linear regression model.

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(c) State the formula for the coefficient of determination R^2 in terms of the quantities `A`, `B` and `C` from the code above.

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